

Ontology Quality Report

Generated automatically from `generate_ontology_report.py`
Date: 2026-03-08

1. Reference Ontologies (TUMedifact)

These two ontologies serve as the baseline for comparison.

Metric	TUMedifact (full)	TUMedifact (trimmed)
Syntax valid	✓	✓
Triples (axioms)	2670	2267
Classes	31	31
Object properties	10	10
Data properties	261	261
Annotation properties	22	9
Max inheritance depth	1	1
Avg inheritance depth	0.8	0.8
Max branching factor	21	21
Avg branching factor	5.2	5.2
Leaf classes	26	26
RR (Relationship Richness)	0.2778	0.2778
AR (Attribute Richness)	8.4194	8.4194
IR (Inheritance Richness)	0.8387	0.8387
Axiom Diversity Score	2	2
Naming: Strict CamelCase	0.9844	0.9870
Naming: Underscore Style	0.0031	0.0000
Naming: Non-conformant	0.0125	0.0130
Label Coverage	0.9564	0.9870
Comment Coverage	0.9408	0.9805

2. Generated Ontologies — Summary by Agent Type

Each folder contains 20 generated ontologies for the EDIFACT domain story.

2.1 High-level Validation Summary

Agent Type	N	Syntax Valid	Hermit Consistent	Pellet Consistent	OOPS Passed	No Critical Pitfalls
workflow	15	100.0%	0.0%	0.0%	100.0%	100.0%
singleAgent	17	100.0%	85.0%	85.0%	100.0%	100.0%
dualAgent	20	100.0%	100.0%	100.0%	100.0%	100.0%
triAgent	11	100.0%	85.0%	70.0%	65.0%	95.0%

2.2 Structural Metrics (avg / median)

Agent Type	Triples avg	Triples med	Classes avg	Classes med	Obj Props avg	Data Props avg	Ann Props avg
workflow	856	845	65.2	65	64.3	48.3	0
singleAgent	325.6	339	18.9	19	28.8	11.4	0.1
dualAgent	316.1	323.5	18.6	18.5	29.5	9.6	0
triAgent	387.9	405	27.3	27	35.6	6.5	0

2.3 Structural Range (min-max)

Agent Type	Triples min	Triples max	Classes min	Classes max
workflow	696	1045	56	74
singleAgent	267	386	16	21
dualAgent	227	373	13	22
triAgent	270	444	22	37

2.4 OOPS Pitfall Summary

Agent Type	Total Pitfall Occurrences	Unique Pitfalls	Most Common
workflow	0	0	None
singleAgent	0	0	None
dualAgent	0	0	None
triAgent	10	5	P04 (×4), P20 (×2), P22 (×2), P12 (×1), P13 (×1)

2.5 Hierarchy Complexity (avg across ontologies in folder)

Agent Type	Max Depth (max)	Max Depth (avg)	Avg Depth (avg)	Max Branching (max)	Avg Branching (avg)	Leaf Classes (avg)
workflow	4	2.5	0.6	39	2.5	50.7
singleAgent	5	1.7	0.4	4	1.6	15.8
dualAgent	4	1.2	0.3	4	1.6	15.7

Agent Type	Max Depth (max)	Max Depth (avg)	Avg Depth (avg)	Max Branching (max)	Avg Branching (avg)	Leaf Classes (avg)
triAgent	3	1.5	0.4	8	2.1	23

2.6 OntoQA Structural Ratios

Agent Type	RR avg	AR avg	IR avg
workflow	0.6000	0.7000	0.5000
singleAgent	0.9000	0.6000	0.3000
dualAgent	0.9000	0.5000	0.3000
triAgent	0.8000	0.2000	0.3000

2.7 Axiom Complexity & Lexical Quality

Agent Type	Axiom Diversity avg	Axiom Diversity max	Naming Strict avg	Naming Underscore avg	Naming Bad avg	Label Coverage avg	Comment Coverage avg
workflow	7.1	8	0.6000	0.4000	0.0000	0.3000	0.1000
singleAgent	4.5	7	1.0000	0.0000	0.0000	1.0000	1.0000
dualAgent	3.4	6	1.0000	0.0000	0.0000	1.0000	1.0000
triAgent	5.2	7	1.0000	0.0000	0.0000	1.0000	1.0000

2.8 Generator Performance

Agent Type	Gen Success Rate	Avg Duration (s)	Avg Iterations	Pitfalls During Gen
workflow	100.0%	332.1	N/A	0
singleAgent	0.0%	0	N/A	0
dualAgent	100.0%	180.1	2.2	9
triAgent	100.0%	1037.9	3.0	0

3. Per-Folder Detail

3.1 workflow

Hermit inconsistent classes (16): `xml_combined_owl.xml.Cl_Information`, `owl.Nothing`, `xml_combined_owl.xml.Cl_ItemSoldDate`, `xml_combined_owl.xml.Cl_Invoice`, `xml_combined_owl.xml.Cl_ItemSKU`, `xml_combined_owl.xml.Cl_ItemName`, `xml_combined_owl.xml.Cl_ItemPrice`, `xml_combined_owl.xml.Cl_InvoiceOrganizationRole`, `xml_combined_owl.xml.Cl_InvoiceLine`, `xml_combined_owl.xml.Cl_ItemImage` ...

Pellet inconsistent classes (70): `xml_combined_owl.xml.Cl_Service`, `xml_combined_owl.xml.Cl_InvoiceLine`, `xml_combined_owl.xml.Cl_OrganizationInvolvementDisplay`, `xml_combined_owl.xml.Cl_ProvidedInformation`, `xml_combined_owl.xml.Cl_RequiredInformation`, `xml_combined_owl.xml.Cl_InvoiceAssignment`, `xml_combined_owl.xml.Cl_SellerRole`, `xml_combined_owl.xml.Cl_DisplayAttribute`, `xml_combined_owl.xml.Cl_InvoiceDetail`, `xml_combined_owl.xml.Cl_OrganizationInformationDisplay` ...

Validation Results

File (15)	Syntax	HermIT	Pellet	OOPS
TUMedifact (trimmed) (baseline)	✓	—	—	—
EDIFACT_combined_turtle_20260301_223728.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_235347.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_223204.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_222612.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_221951.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_215419.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_220409.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_232958.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_231410.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_230354.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_221415.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_233503.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_220919.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_234726.owl	✓	✗	✗	✓
EDIFACT_combined_turtle_20260301_231957.owl	✓	✗	✗	✓

Extracted Details

File (15)	Classes	Obj Props	Data Props	Ann Props	Triples	Max Depth	Avg Depth	Max Branch	Leaves	RR	AR	IR	Axiom Div.	Name Strict	Name Usc.	Name Bad	Label Cov.	Commer Cov.
TUMedifact (trimmed) (baseline)	31	10	261	9	2267	1	0.81	21	26	0.2778	8.4194	0.8387	2	0.987	0.0	0.013	0.987	0.9805
EDIFACT_combined_turtle_20260301_223728.owl	74	75	48	0	899	2	0.51	6	58	0.6696	0.6486	0.5	8	0.6244	0.3756	0.0	0.1421	0.0305
EDIFACT_combined_turtle_20260301_235347.owl	74	69	46	0	1045	4	0.77	8	55	0.6509	0.6216	0.5	6	0.6085	0.3915	0.0	0.5291	0.3862
EDIFACT_combined_turtle_20260301_223204.owl	72	62	46	0	911	2	0.56	7	54	0.62	0.6389	0.5278	8	0.6	0.4	0.0	0.5111	0.2444
EDIFACT_combined_turtle_20260301_222612.owl	69	65	47	0	803	2	0.55	6	53	0.6311	0.6812	0.5507	7	0.5746	0.4254	0.0	0.2044	0.1271
EDIFACT_combined_turtle_20260301_221951.owl	68	58	61	0	882	3	0.71	8	53	0.6105	0.8971	0.5441	7	0.6364	0.3636	0.0	0.262	0.0749
EDIFACT_combined_turtle_20260301_215419.owl	66	53	43	0	713	2	0.62	6	52	0.6235	0.6515	0.4848	6	0.6149	0.3851	0.0	0.087	0.1118
EDIFACT_combined_turtle_20260301_220409.owl	65	67	44	0	845	3	1.06	39	52	0.5194	0.6769	0.9538	7	0.5966	0.4034	0.0	0.125	0.0966
EDIFACT_combined_turtle_20260301_232958.owl	65	62	51	0	816	2	0.43	5	52	0.6813	0.7846	0.4462	6	0.6236	0.3764	0.0	0.2135	0.0281
EDIFACT_combined_turtle_20260301_231410.owl	64	63	52	0	784	2	0.3	3	52	0.7683	0.8125	0.2969	8	0.5866	0.4134	0.0	0.3408	0.1732
EDIFACT_combined_turtle_20260301_230354.owl	63	72	53	0	965	3	0.89	16	48	0.6102	0.8413	0.7302	8	0.633	0.367	0.0	0.4628	0.0106

File (15)	Classes	Obj Props	Data Props	Ann Props	Triples	Max Depth	Avg Depth	Max Branch	Leaves	RR	AR	IR	Axiom Div.	Name Strict	Name Usc.	Name Bad	Label Cov.	Commer Cov.
EDIFACT_combined_turtle_20260301_221415.owl	62	65	47	0	877	2	0.48	7	49	0.6842	0.7581	0.4839	6	0.6416	0.3584	0.0	0.3295	0.0173
EDIFACT_combined_turtle_20260301_233503.owl	62	71	53	0	773	2	0.53	7	52	0.6961	0.8548	0.5	6	0.6667	0.3333	0.0	0.2204	0.0215
EDIFACT_combined_turtle_20260301_220919.owl	59	60	45	0	829	3	0.78	7	42	0.625	0.7627	0.6102	8	0.6402	0.3598	0.0	0.3963	0.0061
EDIFACT_combined_turtle_20260301_234726.owl	59	70	47	0	1002	3	0.97	5	44	0.6667	0.7966	0.5932	8	0.6648	0.3352	0.0	0.6136	0.3352
EDIFACT_combined_turtle_20260301_231957.owl	56	53	41	0	696	2	0.5	10	45	0.6709	0.7321	0.4643	7	0.6267	0.3733	0.0	0.02	0.02

3.2 singleAgent

Hermit inconsistent classes (8): ontology.Buyer, ontology.Organization, ontology.Price, ontology.DeliveryParty, ontology.AgentRole, ontology.OrganizationRoleAssignment, ontology.RoleAssignment, owl.Nothing

Pellet inconsistent classes (8): ontology.Buyer, ontology.Organization, ontology.Price, ontology.DeliveryParty, ontology.AgentRole, ontology.OrganizationRoleAssignment, ontology.RoleAssignment, owl.Nothing

Validation Results

File (17)	Syntax	Hermit	Pellet	OOPS
TUMedifact (trimmed) (baseline)	✓	—	—	—
EDIFACT_ontology_20260304_070119.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_075831.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_062606.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_063345.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_071919.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_072404.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_112519.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_085410.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_090528.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_093001.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_094903.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_071618.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_075320.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_072939.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_092221.owl	✓	✓	✓	✓
EDIFACT_ontology_20260304_111526.owl	✓	✓	✓	✓

File (17)	Syntax	HermiT	Pellet	OOPS
EDIFACT_ontology_20260304_073706.owl	✓	✓	✓	✓

Extracted Details

File (17)	Classes	Obj Props	Data Props	Ann Props	Triples	Max Depth	Avg Depth	Max Branch	Leaves	RR	AR	IR	Axiom Div.	Name Strict	Name Usc.	Name Bad	Label Cov.	Comment Cov.
TUMedifact (trimmed) (baseline)	31	10	261	9	2267	1	0.81	21	26	0.2778	8.4194	0.8387	2	0.987	0.0	0.013	0.987	0.9805
EDIFACT_ontology_20260304_070119.owl	21	32	13	0	345	4	1.29	3	14	0.7442	0.619	0.5238	3	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_075831.owl	21	28	13	1	349	1	0.33	3	17	0.8	0.619	0.3333	6	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_062606.owl	20	28	13	0	348	2	0.5	3	16	0.8235	0.65	0.3	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_063345.owl	20	32	15	0	362	1	0.2	3	18	0.8889	0.75	0.2	4	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_071919.owl	20	22	14	0	285	1	0.3	3	17	0.7857	0.7	0.3	4	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_072404.owl	20	34	13	0	342	1	0.35	3	16	0.8293	0.65	0.35	3	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_112519.owl	20	28	12	0	302	1	0.3	3	16	0.8235	0.6	0.3	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_085410.owl	19	30	9	0	305	1	0.05	1	18	0.9677	0.4737	0.0526	6	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_090528.owl	19	30	11	0	341	1	0.21	3	17	0.8824	0.5789	0.2105	4	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_093001.owl	19	36	14	0	386	5	1.74	4	12	0.7826	0.7368	0.5263	6	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_094903.owl	19	28	15	0	340	2	0.37	4	16	0.8235	0.7895	0.3158	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_071618.owl	18	28	7	0	317	3	0.72	3	14	0.8	0.3889	0.3889	7	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_075320.owl	18	34	6	0	310	1	0.06	1	17	0.9714	0.3333	0.0556	3	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_072939.owl	17	24	8	0	267	1	0.24	3	15	0.8571	0.4706	0.2353	2	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_092221.owl	17	28	14	0	339	2	0.35	3	14	0.8485	0.8235	0.2941	4	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_111526.owl	17	24	9	0	296	1	0.06	1	16	0.96	0.5294	0.0588	3	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260304_073706.owl	16	24	8	0	302	1	0.06	1	15	0.96	0.5	0.0625	7	1.0	0.0	0.0	1.0	1.0

3.3 dualAgent

Validation Results

File (20)	Syntax	HermiT	Pellet	OOPS
TUMedifact (trimmed) (baseline)	✓	—	—	—
EDIFACT_ontology_20260301_222236.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_220437.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_221940.owl	✓	✓	✓	✓

File (20)	Syntax	HermiT	Pellet	OOPS
EDIFACT_ontology_20260301_223501.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_220832.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_224527.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_215924.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_224922.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_225145.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_225409.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_221113.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_221332.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_222534.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_223750.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_225732.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_230018.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_230338.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_215700.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_224212.owl	✓	✓	✓	✓
EDIFACT_ontology_20260301_221725.owl	✓	✓	✓	✓

Extracted Details

File (20)	Classes	Obj Props	Data Props	Ann Props	Triples	Max Depth	Avg Depth	Max Branch	Leaves	RR	AR	IR	Axiom Div.	Name Strict	Name Usc.	Name Bad	Label Cov.	Comment Cov.
TUMedifact (trimmed) (baseline)	31	10	261	9	2267	1	0.81	21	26	0.2778	8.4194	0.8387	2	0.987	0.0	0.013	0.987	0.9805
EDIFACT_ontology_20260301_222236.owl	22	30	12	0	340	1	0.41	4	18	0.7692	0.5455	0.4091	6	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_220437.owl	21	40	3	0	369	2	0.43	3	17	0.8333	0.1429	0.381	3	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_221940.owl	21	28	5	0	308	1	0.33	3	17	0.8	0.2381	0.3333	3	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_223501.owl	21	32	7	0	316	1	0.38	3	17	0.8	0.3333	0.381	2	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_220832.owl	20	38	8	0	352	1	0.3	3	16	0.8636	0.4	0.3	4	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_224527.owl	20	32	13	0	373	1	0.2	3	18	0.8889	0.65	0.2	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_215924.owl	19	34	8	0	322	1	0.11	1	17	0.9444	0.4211	0.1053	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_224922.owl	19	34	7	0	304	1	0.32	3	15	0.85	0.3684	0.3158	2	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_225145.owl	19	30	9	0	308	2	0.47	4	16	0.8108	0.4737	0.3684	4	1.0	0.0	0.0	1.0	1.0

File (20)	Classes	Obj Props	Data Props	Ann Props	Triples	Max Depth	Avg Depth	Max Branch	Leaves	RR	AR	IR	Axiom Div.	Name Strict	Name Usc.	Name Bad	Label Cov.	Comment Cov.
EDIFACT_ontology_20260301_225409.owl	19	30	12	0	315	1	0.11	1	17	0.9375	0.6316	0.1053	2	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_221113.owl	18	28	5	0	261	1	0.28	3	15	0.8485	0.2778	0.2778	2	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_221332.owl	18	32	12	0	327	1	0.06	1	17	0.9697	0.6667	0.0556	2	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_222534.owl	18	30	7	0	348	4	1.06	3	12	0.7895	0.3889	0.4444	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_223750.owl	18	26	15	0	325	1	0.22	3	16	0.8667	0.8333	0.2222	2	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_225732.owl	18	26	16	0	338	1	0.28	3	15	0.8387	0.8889	0.2778	4	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_230018.owl	18	26	14	0	327	1	0.28	3	15	0.8387	0.7778	0.2778	4	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_230338.owl	17	26	14	0	335	1	0.06	1	16	0.963	0.8235	0.0588	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_215700.owl	16	22	9	0	235	1	0.25	3	14	0.8462	0.5625	0.25	2	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_224212.owl	16	24	8	0	292	1	0.25	3	14	0.8571	0.5	0.25	4	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260301_221725.owl	13	22	8	0	227	1	0.08	1	12	0.9565	0.6154	0.0769	2	1.0	0.0	0.0	1.0	1.0

3.4 triAgent

Hermit inconsistent classes (5): ontology.BuyerRole, ontology.GLNIdentifier, ontology.InvoiceMessage, owl.Nothing, ontology.SimpleDataElement

Pellet inconsistent classes (5): ontology.BuyerRole, ontology.GLNIdentifier, ontology.InvoiceMessage, owl.Nothing, ontology.SimpleDataElement

Validation Results

File (11)	Syntax	Hermit	Pellet	OOPS
TUMedifact (trimmed) (baseline)	✓	—	—	—
EDIFACT_ontology_20260302_081000.owl	✓	✓	✓	✓
EDIFACT_ontology_20260302_124024.owl	✓	✓	✓	✗
EDIFACT_ontology_20260303_094621.owl	✓	✓	✓	✗
EDIFACT_ontology_20260302_140235.owl	✓	✓	✓	✓
EDIFACT_ontology_20260302_133553.owl	✓	✓	✓	✓
EDIFACT_ontology_20260302_073654.owl	✓	✓	✓	✓
EDIFACT_ontology_20260303_085821.owl	✓	✓	✓	✗
EDIFACT_ontology_20260303_061520.owl	✓	✓	✓	✓
EDIFACT_ontology_20260303_070100.owl	✓	✓	✓	✓
EDIFACT_ontology_20260302_082612.owl	✓	✓	✓	✓
EDIFACT_ontology_20260303_091850.owl	✓	✓	✓	✓

Extracted Details

File (11)	Classes	Obj Props	Data Props	Ann Props	Triples	Max Depth	Avg Depth	Max Branch	Leaves	RR	AR	IR	Axiom Div.	Name Strict	Name Usc.	Name Bad	Label Cov.	Comment Cov.
TUMedifact (trimmed) (baseline)	31	10	261	9	2267	1	0.81	21	26	0.2778	8.4194	0.8387	2	0.987	0.0	0.013	0.987	0.9805
EDIFACT_ontology_20260302_081000.owl	37	30	8	0	367	2	0.81	8	29	0.5769	0.2162	0.5946	2	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260302_124024.owl	30	27	2	0	314	1	0.3	3	25	0.75	0.0667	0.3	4	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260303_094621.owl	30	42	7	0	440	1	0.37	3	24	0.7925	0.2333	0.3667	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260302_140235.owl	29	40	14	0	426	2	0.38	3	24	0.7843	0.4828	0.3793	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260302_133553.owl	28	23	3	0	270	1	0.25	5	25	0.7667	0.1071	0.25	5	0.9444	0.0556	0.0	1.0	1.0
EDIFACT_ontology_20260302_073654.owl	27	38	2	0	396	3	0.74	3	20	0.76	0.0741	0.4444	7	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260303_085821.owl	27	34	3	0	365	1	0.3	5	24	0.8095	0.1111	0.2963	5	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260303_061520.owl	24	38	11	0	444	1	0.17	2	21	0.9048	0.4583	0.1667	6	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260303_070100.owl	24	34	12	0	411	1	0.21	2	21	0.8718	0.5	0.2083	7	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260302_082612.owl	22	42	5	0	405	2	0.32	3	20	0.913	0.2273	0.1818	6	1.0	0.0	0.0	1.0	1.0
EDIFACT_ontology_20260303_091850.owl	22	44	4	0	429	1	0.27	5	20	0.88	0.1818	0.2727	5	1.0	0.0	0.0	1.0	1.0

4. Comparison with Reference Ontologies

Values compared against **TUMedifact (full)** and **TUMedifact (trimmed)**.

Agent Type	Triples avg	vs Full	vs Trimmed	Classes avg	vs Full	vs Trimmed
workflow	856	-1814	-1411	65.2	+34	+34
singleAgent	325.6	-2344	-1941	18.9	-12	-12
dualAgent	316.1	-2354	-1951	18.6	-12	-12
triAgent	387.9	-2282	-1879	27.3	-4	-4

Reference — *TUMedifact (full)*: 2670 triples, 31 classes

Reference — *TUMedifact (trimmed)*: 2267 triples, 31 classes

5. Key Observations

- **workflow**: HermiT consistency rate is 0.0% — 15 ontologies are inconsistent.
- **workflow**: Average class count is 65.2 (210.3% of reference full ontology's 31 classes).
- **singleAgent**: HermiT consistency rate is 85.0% — 3 ontologies are inconsistent.
- **singleAgent**: Average class count is 18.9 (61.0% of reference full ontology's 31 classes).
- **dualAgent**: Average class count is 18.6 (60.0% of reference full ontology's 31 classes).
- **triAgent**: HermiT consistency rate is 85.0% — 2 ontologies are inconsistent.
- **triAgent**: 10 OOPS pitfall occurrence(s) detected across runs (most common: P04, P20, P22, P12, P13).
- **triAgent**: Average class count is 27.3 (88.1% of reference full ontology's 31 classes).

6. Metric Definitions

Structural Ratios (OntoQA Framework)

- **RR (Relationship Richness)** = $|ObjectProperties| / (|subClassOf\ axioms| + |ObjectProperties|)$. Ratio of non-taxonomic to all relations. Higher values indicate a more interconnected graph rather than a flat taxonomy.
- **AR (Attribute Richness)** = $|DatatypeProperties| / |Classes|$. Average number of data properties per class. Higher values indicate richer per-instance data modelling.
- **IR (Inheritance Richness)** = $|subClassOf\ triples| / |all\ named\ classes|$. Average number of inheritance edges per class (including leaf classes). Distinct from Avg Branching which only considers parent classes.

Axiom Complexity

- **Axiom Diversity Score** (0–10): Count of distinct advanced OWL constructs present in the ontology, out of: `owl:someValuesFrom`, `owl:allValuesFrom`, `owl:disjointWith`, `owl:inverseOf`, `owl:equivalentClass`, `owl:unionOf`, `owl:intersectionOf`, `owl:hasValue`, and cardinality restrictions (`owl:cardinality`, `owl:minCardinality`, `owl:maxCardinality`, counted as one). A score of 0 means the ontology uses only declarations and `rdfs:subClassOf`; a score of 10 means all construct types are present.

Lexical & Annotation Quality

- **Naming: Strict CamelCase** (0.0–1.0): Fraction of named entities matching strict Semantic Web conventions — UpperCamelCase for classes (`^[A-Z][A-Za-z0-9]*$`), lowerCamelCase for properties (`^[a-z][A-Za-z0-9]*$`). No underscores.
- **Naming: Underscore Style** (0.0–1.0): Fraction that follow camelCase with underscores (e.g. `Cl_Invoice`), a common prefix-based variation that would pass if underscores are allowed but failed the strict check.
- **Naming: Non-conformant** (0.0–1.0): Fraction that match neither pattern. The three fractions sum to 1.0.
- **Label Coverage** (0.0–1.0): Fraction of named entities (classes + all property types) that have at least one `rdfs:label` triple. 1.0 = fully labelled.
- **Comment Coverage** (0.0–1.0): Fraction of named entities that have at least one `rdfs:comment` triple. 1.0 = fully documented.

Hierarchy Metrics

- **Max Depth**: Length of the longest `rdfs:subClassOf` chain from a root class (no parent) to a leaf class. Computed via BFS.
- **Avg Depth**: Mean depth across all named classes.
- **Max Branching**: Highest number of direct subclasses any single class has.
- **Avg Branching**: Mean number of direct subclasses per class, computed only over classes that have at least one child (excludes leaf classes).
- **Leaf Classes**: Count of classes with no direct subclasses.